

ACR31511

## Phosphorus oxychloride

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>Phosphorus oxychloride</b><br><b>Phosphorus oxychloride</b>                                                                                                                                                                                                     |
| <b>Cat No. :</b>                                      | <b>315110000</b>                                                                                                                                                                                                                                                   |
| <b>Synonyms</b>                                       | Phosphoryl Chloride                                                                                                                                                                                                                                                |
| <b>CAS No</b>                                         | 10025-87-3                                                                                                                                                                                                                                                         |
| <b>Molecular Formula</b>                              | Cl <sub>3</sub> O P                                                                                                                                                                                                                                                |
| <b>Supplier</b>                                       | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                               |                                |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                                                                                                                                               | <b>Appearance</b><br>Colorless | <b>Odor</b><br>pungent |
| <b>Emergency Overview</b><br>Causes severe skin burns and eye damage. Causes damage to organs through prolonged or repeated exposure. May be corrosive to metals. Harmful if swallowed. Fatal if inhaled. Reacts violently with water. Contact with water liberates toxic gas. Moisture sensitive. Lachrymator (substance which increases the flow of tears). |                                |                        |

#### Classification of the substance or mixture

|                                                      |              |
|------------------------------------------------------|--------------|
| Substances/mixtures corrosive to metal               | Category 1   |
| Acute Oral Toxicity                                  | Category 4   |
| Acute Inhalation Toxicity - Vapors                   | Category 2   |
| Skin Corrosion/Irritation                            | Category 1 A |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| Specific target organ toxicity - (repeated exposure) | Category 1   |

#### Label Elements

## Phosphorus oxychloride



## Signal Word

Danger

## Hazard Statements

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H372 - Causes damage to organs through prolonged or repeated exposure  
H302 - Harmful if swallowed  
H330 - Fatal if inhaled

## Precautionary Statements

## Prevention

P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - Wear respiratory protection

## Response

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P390 - Absorb spillage to prevent material damage  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P402 - Store in a dry place  
P405 - Store locked up  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant liner

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

May be corrosive to metals. Reacts violently with water.

## Health Hazards

Corrosive. Causes skin and eye burns. Causes serious eye damage. Causes damage to organs through prolonged or repeated exposure. Harmful if swallowed. Fatal if inhaled.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Reacts violently with water. Is not likely mobile in the environment. Decomposes in contact with water. Reacts violently with water.

## Other Hazards

Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component              | CAS No     | Weight % |
|------------------------|------------|----------|
| Phosphorus oxychloride | 10025-87-3 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: After inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Powder. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

Contact with water liberates toxic gas. Water.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Contact with water liberates toxic gas. Reacts violently with water.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Should not be released into the environment.

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**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Do not expose spill to water.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere.

**Storage**

Protect from moisture. Corrosives area. Keep under nitrogen. Keep away from water or moist air. Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component              | China                                                     | Taiwan                                      | Thailand     | Hong Kong |
|------------------------|-----------------------------------------------------------|---------------------------------------------|--------------|-----------|
| Phosphorus oxychloride | TWA: 0.3 mg/m <sup>3</sup><br>STEL: 0.6 mg/m <sup>3</sup> | TWA: 0.1 ppm<br>TWA: 0.63 mg/m <sup>3</sup> | TWA: 0.1 ppm | -         |

| Component              | ACGIH TLV    | OSHA PEL                                                       | NIOSH                                                                                    | The United Kingdom                                                                                                 | European Union |
|------------------------|--------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------|
| Phosphorus oxychloride | TWA: 0.1 ppm | (Vacated) TWA: 0.1 ppm<br>(Vacated) TWA: 0.6 mg/m <sup>3</sup> | TWA: 0.1 ppm<br>TWA: 0.6 mg/m <sup>3</sup><br>STEL: 0.5 ppm<br>STEL: 3 mg/m <sup>3</sup> | STEL: 0.6 ppm 15 min<br>STEL: 3.8 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 ppm 8 hr<br>TWA: 1.3 mg/m <sup>3</sup> 8 hr |                |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists  
OSHA - Occupational Safety and Health Administration  
NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use only under a chemical fume hood. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles Face protection shield (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Butyl rubber   | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger

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of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                             |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387                                       |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                             |                                          |
|------------------------------------------------|-----------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                   |                                          |
| <b>Physical State</b>                          | Liquid                      |                                          |
| <b>Odor</b>                                    | pungent                     |                                          |
| <b>Odor Threshold</b>                          | No data available           |                                          |
| <b>pH</b>                                      | No information available    |                                          |
| <b>Melting Point/Range</b>                     | 1.2 °C / 34.2 °F            |                                          |
| <b>Softening Point</b>                         | No data available           |                                          |
| <b>Boiling Point/Range</b>                     | 107 °C / 224.6 °F           |                                          |
| <b>Flash Point</b>                             | No information available    | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available           |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable              | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available           |                                          |
| <b>Vapor Pressure</b>                          | 36 mbar @ 20 °C             |                                          |
| <b>Vapor Density</b>                           | 5.3                         | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.645                       |                                          |
| <b>Bulk Density</b>                            | Not applicable              | Liquid                                   |
| <b>Water Solubility</b>                        | Reacts violently with water |                                          |
| <b>Solubility in other solvents</b>            | No information available    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                             |                                          |
| <b>Autoignition Temperature</b>                | No data available           |                                          |
| <b>Decomposition Temperature</b>               | No data available           |                                          |
| <b>Viscosity</b>                               | 1.11 mPa.s at 22 °C         |                                          |
| <b>Explosive Properties</b>                    | No information available    |                                          |
| <b>Oxidizing Properties</b>                    | No information available    |                                          |
| <b>Molecular Formula</b>                       | Cl <sub>3</sub> O P         |                                          |
| <b>Molecular Weight</b>                        | 153.33                      |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Reacts violently with water. Moisture sensitive. Contact with water liberates toxic gas.  |
| <b>Hazardous Reactions</b>              | Reacts violently with water.                                                              |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                  |
| <b>Conditions to Avoid</b>              | Excess heat. Incompatible products. Exposure to moist air or water. Exposure to moisture. |
| <b>Materials to avoid</b>               | Strong bases. Alcohols. Amines. Metals. Acids. Reducing Agent. Water. Oxidizing agent.    |
| <b>Hazardous Decomposition Products</b> | Oxides of phosphorus. Hydrogen chloride gas.                                              |

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component              | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation                          |
|------------------------|--------------------------|-----------------------------|------------------------------------------|
| Phosphorus oxychloride | LD50 = 380 mg/kg ( Rat ) | LD50 > 250 mg/kg ( Rabbit ) | LC50 = 308 mg/m <sup>3</sup> ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Category 1 A

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

**Respiratory**  
**Skin**

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;** Category 1

**Target Organs**

Respiratory system.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Other Adverse Effects****Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: After inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed

**SECTION 12. ECOLOGICAL INFORMATION**

|                                              |                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>                   | Reacts with water so no ecotoxicity data for the substance is available.                              |
| <b>Persistence and Degradability</b>         |                                                                                                       |
| <b>Persistence</b>                           | Persistence is unlikely, based on information available.                                              |
| <b>Degradability</b>                         | Decomposes in contact with water, Reacts with water.                                                  |
| <b>Degradation in sewage treatment plant</b> | Decomposes in contact with water. Reacts violently with water.                                        |
| <b>Bioaccumulative Potential</b>             | Product does not bioaccumulate due to reaction with water                                             |
| <b>Mobility in soil</b>                      | Decomposes in contact with water. Reacts violently with water Is not likely mobile in the environment |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors                             |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance                                        |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance                                        |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                                            |                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                          |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                       |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge. |

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                                |                        |
|--------------------------------|------------------------|
| <b>UN-No</b>                   | UN1810                 |
| <b>Proper Shipping Name</b>    | PHOSPHORUS OXYCHLORIDE |
| <b>Hazard Class</b>            | 6.1                    |
| <b>Subsidiary Hazard Class</b> | 8                      |
| <b>Packing Group</b>           | I                      |

**IMDG/IMO**

|                                |                        |
|--------------------------------|------------------------|
| <b>UN-No</b>                   | UN1810                 |
| <b>Proper Shipping Name</b>    | PHOSPHORUS OXYCHLORIDE |
| <b>Hazard Class</b>            | 6.1                    |
| <b>Subsidiary Hazard Class</b> | 8                      |
| <b>Packing Group</b>           | I                      |

**IATA**

FORBIDDEN FOR IATA TRANSPORT

|                                |                                                      |
|--------------------------------|------------------------------------------------------|
| <b>UN-No</b>                   | UN1810                                               |
| <b>Proper Shipping Name</b>    | PHOSPHORUS OXYCHLORIDE, FORBIDDEN FOR IATA TRANSPORT |
| <b>Hazard Class</b>            | 6.1                                                  |
| <b>Subsidiary Hazard Class</b> | 8                                                    |
| <b>Packing Group</b>           | I                                                    |

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component              | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Phosphorus oxychloride | X                                                   | X                                       | X    | X     | 233-046-7 | X    | X   | X     | X    | X    | X    | KE-28728 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 29-Oct-2006  
Revision Date 05-Apr-2024  
Revision Summary Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**VPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)



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**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**